GaitCam camera calibration · 相机校准

Two targets. The lens board corrects the 130° lens distortion (once per camera). The floor strips give the floor plane and the centimetre scale (whenever the camera or walkway moves).
两种图案：镜头板校正 130° 镜头的畸变（每台相机一次）；地板纸条给出地板平面和厘米换算（相机或走道移动后重做）。

1 Print 打印

Print pages 2 and 3 at 100% / “Actual size” — never “Fit to page”. Measure the 100 mm bar on each page with a ruler: it must be 100 mm (±0.5 mm), otherwise fix the printer scaling and print again. Laser printer, matte paper (no glossy paper: glare).

第 2、3 页用 100% /“实际大小”打印，千万不要选“适合页面”。用尺子量每页的 100 mm 刻度条，必须是 100 mm（±0.5 mm），否则调整打印缩放后重打。用激光打印机、哑光纸（不要光面纸，会反光）。

2 Lens board (page 2) — once per camera 镜头标定板（第 2 页）—— 每台相机做一次

a. Focus the lens on the walkway floor, then tighten the lens lock screw. Do not turn the lens afterwards; if it is turned, repeat this step.

a. 先把镜头焦距对准走道地板，再拧紧镜头锁定螺丝。之后不要再转镜头；转过就要重做这一步。

b. Glue or tape page 2 completely flat onto a stiff board (foam board, clipboard). A bent sheet ruins the result.

b. 把第 2 页完全平整地贴在硬板上（泡沫板、写字夹板）。纸一弯，结果就不准。

c. The camera must see the board directly, not through the frosted floor: take the floor off for this step (or do it before the camera is mounted).

c. 相机要直接看到板子，不能隔着磨砂底板：做这一步时把底板拿掉（或在装相机之前做）。

d. In GaitCam type the animal ID LENS and press Record. For about 60 seconds move the board slowly 10–20 cm in front of the camera: the centre, all four corners and all four edges of the picture, tilted up to about 45° in different directions. Move slowly and pause now and then. The board may be partly out of the picture.

d. 在 GaitCam 里把编号填 LENS，点开始录制。约 60 秒内，把板子放在相机前 10–20 cm 慢慢移动：画面中间、四个角、四条边都要覆盖到，并朝不同方向倾斜到约 45°。慢慢移，时不时停一下。板子部分出画没关系。

3 Floor strips (page 3) — every time the camera or the walkway is moved 地板纸条（第 3 页）—— 相机或走道每次移动后

a. Cut out the three strips along the dashed lines (keep the white border).

a. 沿虚线剪下三条纸条（保留白边）。

b. With the floor in place, lay the strips printed side DOWN on the walkway floor, end to end along the walkway, centred, pressed flat.

b. 装好底板，把纸条印刷面朝下平放在走道地板上，沿走道首尾相接、居中、压平。

c. Switch on the usual walkway light and click “Record a 1-s calibration clip” in GaitCam. Take the strips away before the animals walk.

c. 打开平时用的走道灯，在 GaitCam 里点“录 1 秒标定片段”。放小鼠之前把纸条拿走。

4 Keep the files 保存文件

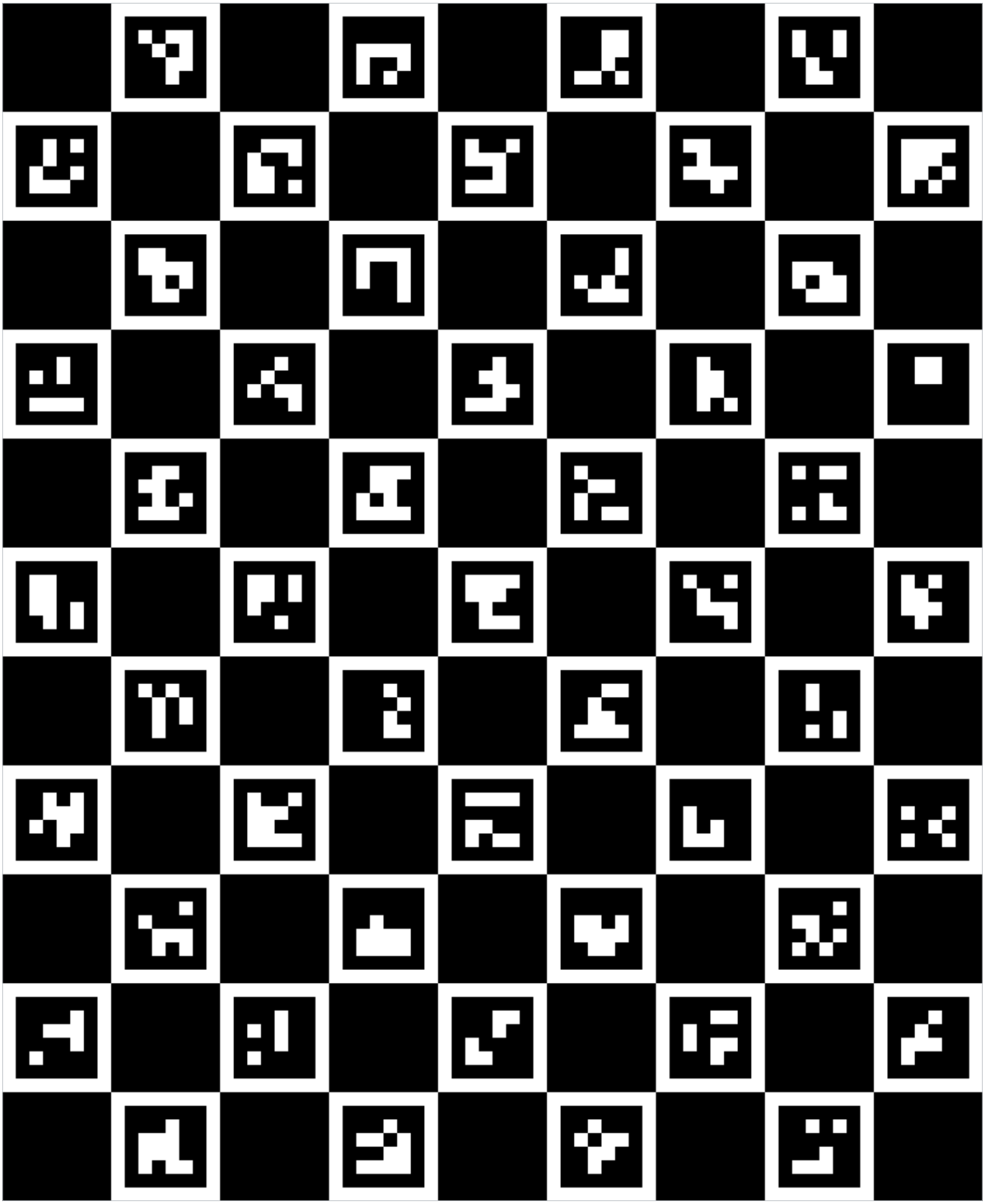
That day's data folder now contains LENS_t01_⋯.mkv and the calib folder. Keep them with the videos: the lens model, the floor mapping and the centimetre scale for those videos are computed from them.

当天的数据文件夹里会有 LENS_t01_⋯.mkv 和 calib 文件夹，请和视频一起保存：这些视频的镜头模型、地板映射和厘米换算都由它们计算。

Lens board: ChArUco 9×11 squares, 20.0 mm squares, 15.0 mm markers, OpenCV DICT_4X4_50. Floor strip: checkerboard 4×24 squares of 10.0 mm (OpenCV pattern 23×3).
Letter paper · generated by GaitCam calibration/make_targets.py

GaitCam lens calibration board ・ 镜头标定板

ChArUco 9×11 squares ・ square 20.0 mm ・ marker 15.0 mm ・ OpenCV DICT_4X4_50 — mount flat on a stiff board / 平整贴在硬板上



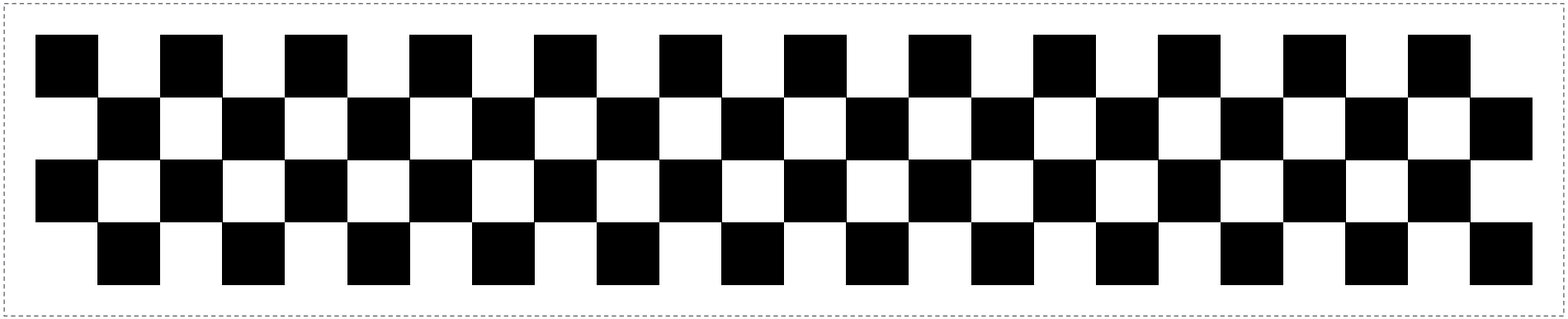
100 mm — must measure exactly 100 mm with a ruler / 用尺子量必须正好 100 mm

Print at 100% / Actual size — not “Fit to page”. 打印选 100% /“实际大小”，不要“适合页面”。

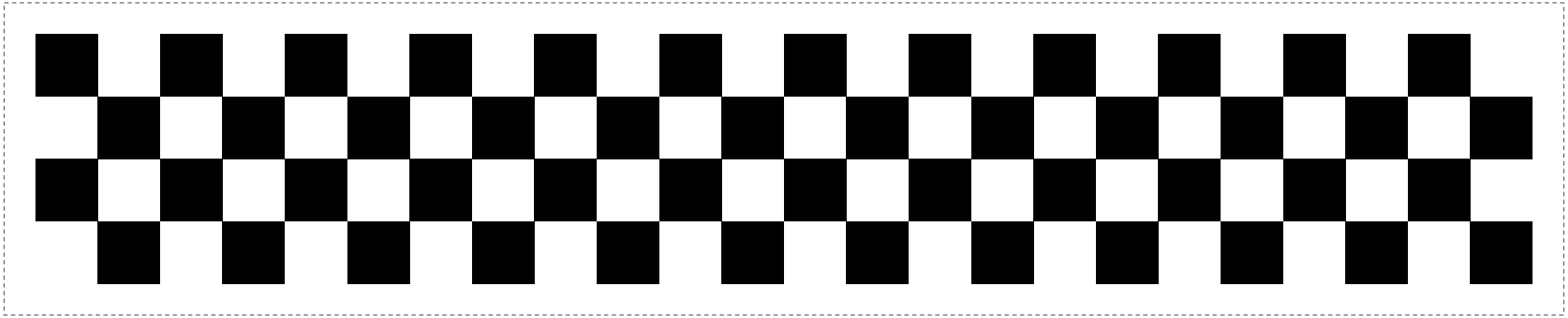
Letter paper

GaitCam floor strips · 地板纸条

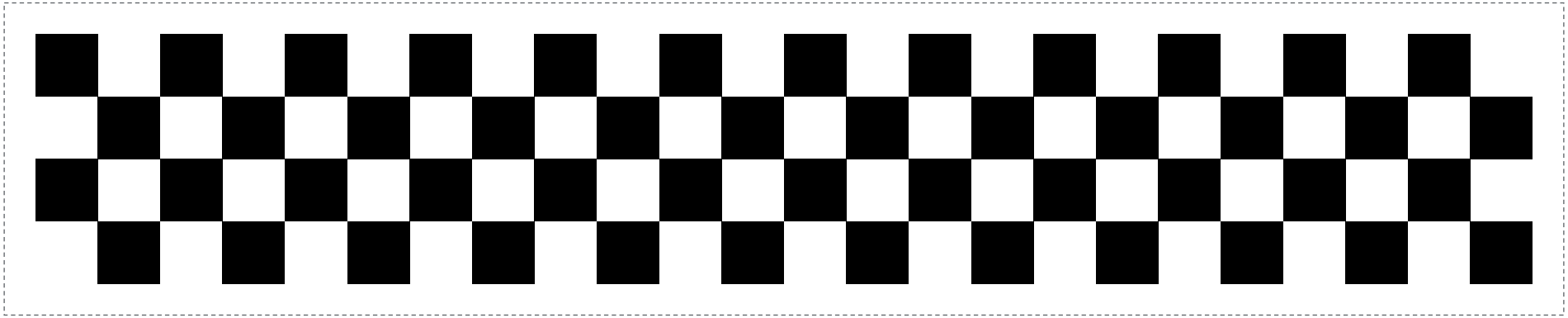
Checkerboard 4×24 squares · 10.0 mm (OpenCV pattern 23×3) — cut along the dashed line, lay printed side DOWN on the walkway floor / 沿虚线剪下，印刷面朝下平放在走道地板上



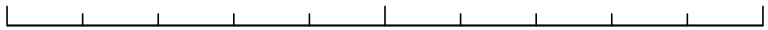
strip A



strip B



strip C



100 mm — must measure exactly 100 mm with a ruler / 用尺子量必须正好 100 mm

Print at 100% / Actual size. 打印选 100% /“实际大小”。 (Letter)